

Dharani Vanukuri

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EXPERIENCE

Data Analyst Intern

University of North Texas

January 2025 – Present

Denton, Texas

- Engineered automated ETL pipeline using Python and SQL to extract, transform, and load 2M+ survey responses from external data sources into a normalized data warehouse, reducing manual processing time by 85% and enabling real-time analytics on library resource utilization.
- Developed machine learning classification model using scikit-learn and regex-based feature engineering to automatically match and categorize 500K+ survey URLs to licensed library resources with 92% accuracy, eliminating 200+ hours of annual manual data matching.
- Built interactive data visualization dashboards in Tableau and Power BI analyzing correlation between library resource usage and faculty research output, delivering actionable insights to senior stakeholders that directly informed \$2M+ resource allocation decisions.
- Implemented scalable, reusable data transformation modules using Python (pandas, NumPy) and SQL to clean, normalize, and parse complex URL datasets across multiple sources, improving data quality by 40% and establishing standardized operational processes adopted department-wide.

Data Engineer

Tiger Analytics

June 2023 – July 2024

Chennai, India

- Designed and maintained end-to-end ETL pipelines using Python, PySpark, and AWS Glue to ingest, transform, and load 50M+ records daily from 12+ heterogeneous sources (REST APIs, relational databases, flat files) into a centralized Redshift data warehouse, reducing data latency from 24 hours to under 2 hours and achieving 99.6% SLA compliance.
- Built scalable, self-service reporting solutions using AWS S3, Athena, and Redshift Spectrum with parameterized SQL views that empowered 30+ finance and business stakeholders to independently query data, reducing ad-hoc request volume by 65% and eliminating 15+ hours of manual reporting per sprint.
- Developed optimized dimensional data models (star and snowflake schemas with SCD Type 2) in Amazon Redshift and implemented automated data quality monitoring using Python and CloudWatch to validate schema integrity, null rates, and row-count thresholds across 40+ production datasets, improving query performance by 70%.
- Collaborated with cross-functional finance and operations teams to translate business requirements into technical specifications using Amazon's working-backward methodology, delivering 8 data products supporting revenue forecasting, and engineered a real-time streaming pipeline using Kafka and AWS Lambda to process transactional event data informing pricing decisions impacting \$5M+ in annual revenue.

EDUCATION

Master of Science in Information Systems and Technology

University of North Texas

Aug 2024 - Present

Denton, Texas

Bachelor of Technology in Computer Science and Engineering

Vellore Institute of Technology, AP

2020 - 2024

AP, India

PROJECTS

Enterprise RAG-Powered Knowledge Assistant using LLMs | Python, LangChain, OpenAI, FAISS, AWS Jan 2026

- Architected a Retrieval-Augmented Generation (RAG) system using LangChain, OpenAI GPT-4 API, and FAISS vector database to enable natural language querying over 100K+ unstructured enterprise documents (PDFs, Confluence pages, Slack exports), achieving 94% answer relevance score on domain-specific evaluation benchmarks.
- Engineered a multi-stage document ingestion pipeline using Python and LangChain text splitters to chunk, embed (OpenAI Ada-002), and index 500K+ text segments into FAISS with metadata filtering, reducing retrieval latency to under 300ms and supporting real-time conversational queries.
- Implemented advanced prompt engineering techniques including chain-of-thought reasoning, few-shot examples, and dynamic context window management to reduce hallucination rate by 38%, and built a guardrail layer using semantic similarity thresholds to detect and flag low-confidence responses before surfacing to end users.
- Deployed the full-stack application using FastAPI, Streamlit, and AWS (EC2, S3, Lambda) with CI/CD via GitHub Actions, serving 200+ daily active users across engineering and product teams and reducing average internal knowledge retrieval time from 25 minutes to under 45 seconds.

Scalable Multi-Source Data Warehouse and Analytics Platform | Python, AWS, Airflow, SQL

Oct 2025

- Designed and deployed a cloud-native data warehouse on AWS Redshift integrating data from 8+ sources (REST APIs, relational databases, flat files) using Python-based ETL orchestration with Apache Airflow, processing 20M+ records across daily scheduled DAGs with 99.5% pipeline uptime.

- Implemented dimensional data models (star schema) with slowly changing dimensions (SCD Type 2) to maintain historical accuracy across customer, product, and transaction datasets, enabling time-series trend analysis for business intelligence use cases.
- Built automated data quality validation framework using Great Expectations and SQL-based assertion checks, scanning 100% of ingested records for schema violations, duplicate detection, and referential integrity — reducing downstream data incidents by 75%.
- Created parameterized, self-service SQL views and materialized tables in Redshift Spectrum that allowed non-technical finance stakeholders to independently generate weekly KPI reports, eliminating 15+ hours of manual reporting effort per sprint cycle.

TECHNICAL SKILLS

Programming Languages: Python, SQL, PySpark, Scala, Java, HiveQL, SparkSQL

Databases & Warehousing: Amazon Redshift, MySQL, PostgreSQL, FAISS, Data Modeling (Star/Snowflake Schema, SCD Type 2)

ETL & Orchestration: Apache Airflow, AWS Glue, Apache Kafka, ETL Pipeline Design, Great Expectations

Cloud & Infrastructure: AWS (S3, Redshift, Athena, Lambda, EC2, Glue, SageMaker, CloudWatch), CI/CD (GitHub Actions)

AI/ML & GenAI: LangChain, OpenAI API, RAG, FAISS, Scikit-learn, TensorFlow, SHAP, Prompt Engineering

Visualization & Reporting: Tableau, Power BI, Matplotlib, Streamlit

Tools & Frameworks: FastAPI, Git, Jupyter Notebook, Pandas, NumPy, Docker, Google Colab